Causality - SoSe 2025

Covariate Adjustment

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Last Time

In the last lecture, we discussed the following concepts:

- Structural Causal Models
- *do* Interventions
- Causal Bayesian Networks
- Average Treatment Effect

Last Time

Structural Causal Models

Is the following a minimal structural causal model? Assume that all noise terms are pairwise independent and follow a standard normal distribution.

$$
\begin{aligned}
Z &= N_Z \\
X &= Z + N_X \\
Y &= 2Z + 0X + N_Y
\end{aligned}
$$

Last Time

Structural Causal Models

Is the following a minimal structural causal model? Assume that all noise terms are pairwise independent and follow a standard normal distribution.

$$Z = N_Z$$
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Last Time

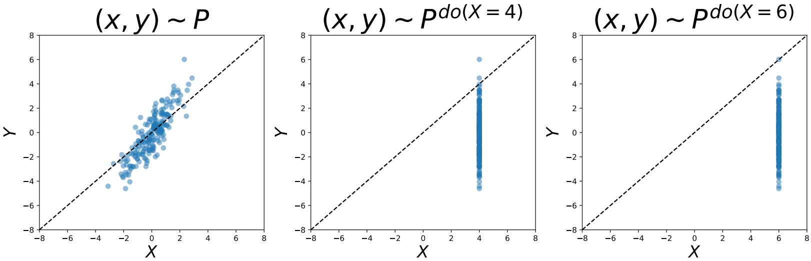
Structural Causal Models

Is the following a minimal structural causal model? Assume that all noise terms are pairwise independent and follow a standard normal distribution.

$$Z := N_Z$$
$$X := Z + N_X$$
$$Y := 2Z + X + N_Y$$

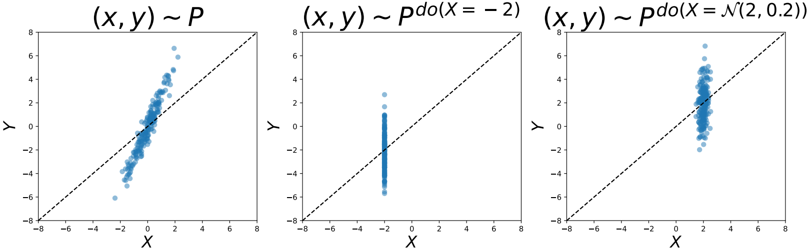
What graph does it generate?

Last Time



Consider the data above. Can we say that Y has a *causal effect* on X or vice versa? What would be a possible graph to generate the data?

Last Time



Consider the data above. Can we say that Y has a *causal effect* on X or vice versa? What would be a possible graph to generate the data?

Last Time

The pair $(\mathcal{G}, P)$ is a **causal Bayesian network** if for any $\boldsymbol{W} \subset \boldsymbol{V}$

$$\underbrace{p(x_{\boldsymbol{V}} | do(X_{\boldsymbol{W}} = x_{\boldsymbol{w}}))}_{\substack{\text{post-interventional distribution} \\ \text{needed to define causal effects}}} = \underbrace{\prod_{j \in \boldsymbol{V} \setminus \boldsymbol{W}} p(x_j | x_{\mathrm{Pa}(j)})[\boldsymbol{X}_{\boldsymbol{W}} = \boldsymbol{x}_{\boldsymbol{w}}]}_{\substack{\text{conditional distribution that can} \\ \text{be computed from observational data}}} \; .$$

Causal DAGs imply strong assumptions, allowing us to estimate *post-intervention distributions* from *observational data*.

Outline

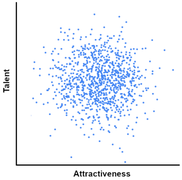
In today's lecture, we will learn *how to adjust for confounding*. In particular, we discuss the following concepts:

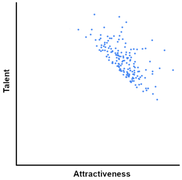
- Confounding and Selection Bias
- Path Method for Linear SCMs
- Adjustment Sets
- Adjustment for Direct Causes
- Identifiability
- Back-Door Criterion

Selection Bias

Example of Berkson's paradox:

- Assume that **attractiveness** and **talent** are independent features (top).
- Now, consider that we **only observe celebrities** in our dataset (bottom).
- A person must be **both talented and attractive**, in some combination, to have become a celebrity.
- As a result, **attractiveness** and **talent** are **negatively correlated** in this **selected** dataset.

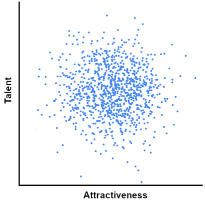




[1] Source: Wikipedia (https://en.wikipedia.org/wiki/Berkson%27s_paradox)

Causal View on Berkson's Paradox

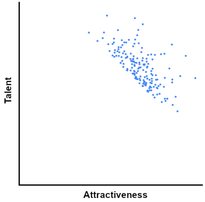
Berkson's paradox is also known as **collider bias**, since it can be modelled through a **collider structure** "$\rightarrow C \leftarrow$" in a DAG.



Here, we assume that being a **celebrity** (C) is a **function** of **attractiveness** (A) and **talent** (T):

$$C := f(A, T, \epsilon)$$

with respect to the DAG $A \rightarrow C \leftarrow T$.



Selection Bias

- Alternatively, **collider bias** is also known as **selection bias**.
- Intuitively, we only observe a dataset which is **pre-selected** with respect to some **selection criterion**.

Different from **observed data**, where we assume that we have access to a distribution $P(\cdot)$ over variables V, in the **selection bias** setting, we only observe data drawn from the **conditional distribution**

$$P(\cdot \mid S = 1) \; ,$$

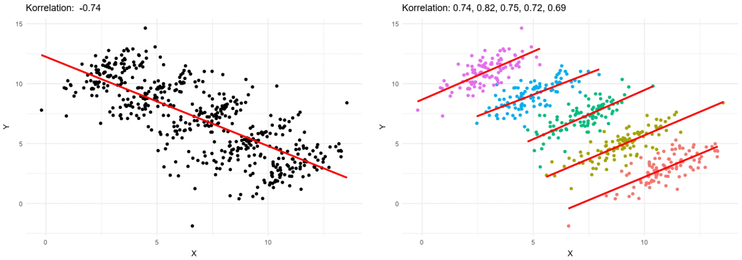
where S is the **selection variable.**

Thus, we can see that **conditioning on a collider** can *induce a dependence*. For example, by conditioning on *Celebrity*, $A \not\perp\!\!\!\perp T | C$.

Confounding

Confounding Continuous Data[2]

The data below has been generated through a confounder ($X \leftarrow C \rightarrow Y$). As we can see, taking the third variable into account (right) explains the data much better than just looking at X and Y (left).



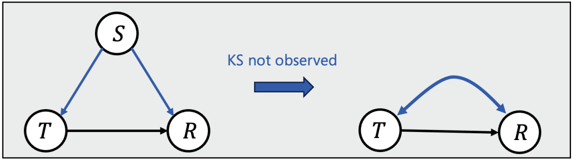
[2]Source: Wikipedia (https://en.wikipedia.org/wiki/Simpson's_paradox)

Confounding (Kidney Stones)

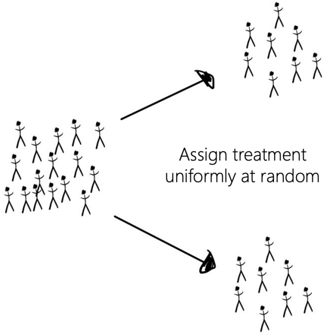
	Overall	Patients with small stones	Patients with large stones
Treatment *a*: Open surgery	78% (273/350)	**93%** (81/87)	**73%** (192/263)
Treatment *b*: Percutaneous nephrolithotomy	**83%** (289/350)	87% (234/270)	69% (55/80)

Figure 1: Simpson's paradox: Recovery rate after different treatments for kidney stones

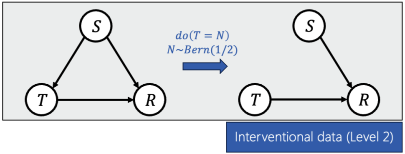
Causal view: (Unobserved) **confounding** through kidney stone size:



Randomized Controlled Trials (Interventional Data)



RCT as an **intervention** on the data generative process



Such data allows for computing, e.g., **average treatment effects**:

$$ATE = \mathrm{E}[R \mid do(T = a)] - \mathrm{E}[R \mid do(T = b)]$$
$$\overset{\text{RCT}}{=} \mathrm{E}[R \mid T = a] - \mathrm{E}[R \mid T = b]$$

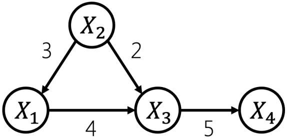
How can we use CBNs for computing post-interventional distributions from observational data?

Total Causal Effects for Linear SCMs

Question: How can we compute the **total average causal effect** for the following **linear SCM**?

SCM: All $N_i \sim \mathcal{N}(0,1)$, $N_i \perp\!\!\!\perp N_j$ for $i \neq j$,

$$X_2 := N_2$$
$$X_1 := 3X_2 + N_1$$
$$X_3 := 4X_1 + 2X_2 + N_3$$
$$X_4 := 5X_3 + N_4$$



- **Direct average causal effects** are the edge weights (linear SCM)
- What is the **total average causal effect** of X_1 on X_4?
  - Increasing X_1 by 1 will on average increase X_3 by $4 \cdot 1 = 4$
  - Increasing X_3 by 4 will on average increase X_4 by $4 \cdot 5 = 20$

Simulating Interventional Distributions

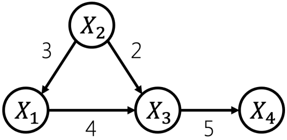
Recall that we can obtain the total average causal effect from **interventional distributions**. Let's simulate them!

Python Code for SCM:

```
import numpy as np
import matplotlib.pyplot as plt
np.random.seed(42)

# Generate external sources
n = 1000
N_1 = np.random.normal(0, 1, n)
N_2 = np.random.normal(0, 1, n)
N_3 = np.random.normal(0, 1, n)
N_4 = np.random.normal(0, 1, n)

# Generate linear additive SCM
X_2 = N_2
X_1 = 3*X_2 + N_1
X_3 = 2*X_2 + 4*X_1 + N_3
X_4 = 5*X_3 + N_4
```



Simulating Interventional Distributions

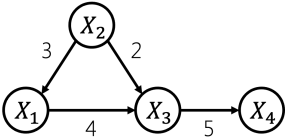
Recall that we can obtain the total average causal effect from **interventional distributions**. Let's simulate them!

Python Code for Interventions:

```
X_2 = N_2
X_1 = np.full(n, 1) # do(X_1 = 1)
X_3 = 2*X_2 + 4*X_1 + N_3
X_4 = 5*X_3 + N_4
m1 = np.mean(X_4)
print(m1)           # E[X_4 | do(X_1 = 1)]

X_2 = N_2
X_1 = np.full(n, 0) # do(X_1 = 0)
X_3 = 2*X_2 + 4*X_1 + N_3
X_4 = 5*X_3 + N_4
m0 = np.mean(X_4)
print(m0)           # E[X_4 | do(X_1 = 0)]
print(m1 - m0)      # ACE
```

```
20.718814227484906
0.7188142274849072
20.0
```



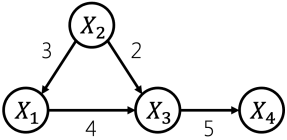
The Path Method for Linear SCMs

Definition (Path Method) Compute the total causal effect of X_i on X_j in a linear SCM as follows:

- For each **directed path** from i to j, multiply the edge weights along the path
- Sum up the results over all paths

There is only a single directed path from 1 to 4, i.e., $(1, 3, 4)$. We again confirm that:

$$4 \cdot 5 = 20 = ACE$$



Path Method (Sample Version)

```
from sklearn.linear_model import LinearRegression

model1 = LinearRegression()
model1.fit(np.column_stack((X_1, X_2)), X_3)
print("Intercept:", model1.intercept_)
print("Coefficients:", model1.coef_)

model2 = LinearRegression()
model2.fit(X_3.reshape(-1,1), X_4)
print("Intercept:", model2.intercept_)
print("Coefficients:", model2.coef_)
print("Estimated ACE:", model1.coef_[0]*model2.coef_[0])
```

```
Intercept: 0.006133587796827156
Coefficients: [4.02180582 1.92440519]
Intercept: -0.01447532205509372
Coefficients: [4.99605171]
Estimated ACE: 20.093149864182003
```

Note that we need to include X_2 in the regression $X_3 \sim X_1, X_2$. In other words, we had to **adjust** for X_2.

Adjustment Sets

Adjustment Sets

Question: Which adjustment sets S are valid?

$S = \emptyset$

```
model = LinearRegression()
print("Estimated ACE:", model.fit(X_1.reshape(-1,1), X_4).coef_[0])  # invalid
```

```
Estimated ACE: 23.01902338507625
```

$S = \{X_2\}$

```
print("Est. ACE:", model.fit(np.column_stack((X_1, X_2)), X_4).coef_[0])  # valid
```

```
Est. ACE: 20.092711607960705
```

$S = \{X_3\}$

```
print("Est. ACE:", model.fit(np.column_stack((X_1, X_3)), X_4).coef_[0])  # invalid
```

```
Est. ACE: -0.08801369434007886
```

$S = \{X_2, X_3\}$

```
print("Est. ACE:", model.fit(np.column_stack((X_1, X_2, X_3)), X_4).coef_[0])  # invalid
```

```
Est. ACE: -0.10588543364615033
```

Determining Adjustment Sets

Let $(\mathcal{G}, P)$ be a causal Bayesian network, $(i, k) \in \boldsymbol{V}, i \neq k$. Can we find a **graphical criterion** for sets $\boldsymbol{Z} \subset \boldsymbol{V}$ that satisfy

$$p(x_k | do(x_i)) = \int_{x_{\boldsymbol{Z}}} p(x_k | x_i, x_{\boldsymbol{Z}}) p(x_{\boldsymbol{Z}}) dx_{\boldsymbol{Z}} \tag{1}$$

for all $p(\cdot)$ such that $(\mathcal{G}, P)$ is a CBN?

- Eq. (1) is called the **adjustment formula**
- Set $\boldsymbol{Z}$ satisfying Eq. (1) are called **valid adjustment sets**

Determining Adjustment Sets

Question: How can we find valid adjustment sets?

- Should we adjust for as many variables as possible?
  - We already saw that this can lead to errors; additionally, it can decrease the statistical efficiency
- Is it always safe to adjust for "pre-treatment" variables?

No Direct Causes

Let $(\mathcal{G}, P)$ be a CBN, and assume there are **no edges into** X_i, i.e., $\mathrm{Pa}(i) = \emptyset$. Then it follows from the **truncated factorization formula** that

$$\begin{aligned} p(x_{\boldsymbol{V}\setminus\{i\}} | do(X_i = x_i')) &= \prod_{j \in \boldsymbol{V}\setminus\{i\}} p(x_j | x_{\mathrm{Pa}(j)})\Big|_{x_i'} \\ &= \frac{p(x_{\boldsymbol{V}})}{p(x_i)}\bigg|_{x_i'} = p(x_{\boldsymbol{V}\setminus\{i\}} | x_i') \end{aligned}$$

where $p(x_i) = p(x_i | x_{\mathrm{Pa}(i)})$ since $\mathrm{Pa}(i) = \emptyset$.

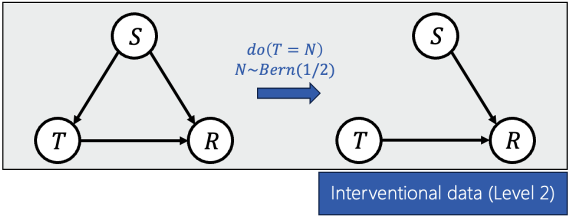
In this special case, the **do-operator** is equal to **regular conditioning**!

Further, for any $k \neq i$, integrating out the other variables yields

$$p(x_k | do(x_i')) = p(x_k | x_i')$$

No Direct Causes

The situation with $\mathrm{Pa}(i) = \emptyset$ arises in a randomized controlled trial, where $T = X_i$ is determined by a coin toss:



Reweighting

Let $(\mathcal{G}, P)$ be a CBN, it generally holds that

$$p(x_{\boldsymbol{V}\setminus\{i\}} | do(X_i = x_i')) = \prod_{j \in \boldsymbol{V}\setminus\{i\}} p(x_j | x_{\mathrm{Pa}(j)})\Big|_{x_i'} = \frac{p(x_{\boldsymbol{V}})}{p(x_i | x_{\mathrm{Pa}(i)})}\Bigg|_{x_i'}$$

- Thus, the interventional distribution is a **re-weighted** version of the observational distribution, using weights $1/p(x_i | x_{\mathrm{Pa}(i)})$
- This is used in **inverse probability weighting** (IPW) in marginal structural models (Robins, Hernan, Brumback)

Adjusting for Direct Causes

Let $(\mathcal{G}, P)$ be a CBN, it generally holds that

$$p(x_{\mathbf{V}\setminus\{i\}} | do(X_i = x_i)) = \frac{p(x_{\mathbf{V}})}{p(x_i | x_{\mathrm{Pa}(i)})} = p(x_{\mathbf{V}\setminus\{i,\mathrm{Pa}(i)\}} | x_i, x_{\mathrm{Pa}(i)}) p(x_{\mathrm{Pa}(i)})$$

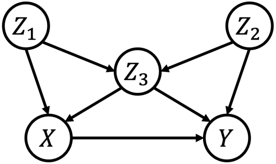
Lemma Let $k \notin \{i, \mathrm{Pa}(i)\}$, then integrating out all variables other than X_i and X_k yields

$$p(x_k | do(x_i)) = \int_{x_{\mathrm{Pa}(i)}} p(x_k | x_i, x_{\mathrm{Pa}(i)}) p(x_{\mathrm{Pa}(i)}) dx_{\mathrm{Pa}(i)} \ ,$$

which is known as **adjusting for $X_{\mathrm{Pa}(i)}$**.

Example

Question: Can we compute the causal effect of X on Y in this graph?

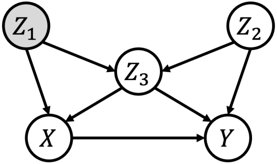


We need to compute $p(y|do(x))$, which we can do by:

- Truncated factorization formula
- Parent adjustment with adjustment variable $Z = \{Z_1, Z_3\}$

Example

Question: Can we compute the causal effect of X on Y in this graph, where Z_1 (gray) is unobserved?



In other words: Can we compute $p(y|do(x))$ if (only) Z_1 is not measured?

- That is, is $p(y|do(x))$ **identifiable** if (only) Z_1 is not measured?

Identifiability

Identifiability

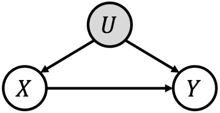
Intuition: An aspect of a statistical model is **identifiable** when it cannot be changed without there also being some change in the distribution of the observable variables.

- If we can alter part of a model with no observable consequences, that part of the model is **unidentifiable**
- Identification is about the true distribution, not about finite data

Identifiability

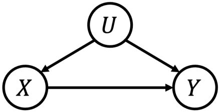
Unobserved Common Cause

- $p(y|x)$ is identifiable
- $p(y|do(x))$ is **not identifiable**: can have **different** $p(y|do(x))$ with **same** observable distribution $p(x, y)$
- U could compensate changes in X
- Cannot estimate $p(y|do(x))$ from observations



Observed Common Cause

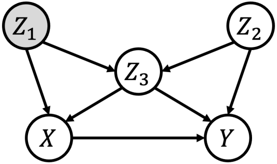
- Can write $p(y|do(x))$ in terms of distributions of observables
- Confounding can be removed by an **identification strategy**
- $p(y|do(x))$ is **identifiable**



Back-Door Criterion

Example

Question: Can we compute the causal effect of X on Y in this graph, where Z_1 (gray) is unobserved?



Yes, $\{Z_2, Z_3\}$ fulfills the back-door criterion!

Back-Door Criterion (Pearl)

Definition Let $(\mathcal{G}, P)$ be a causal Bayesian network, $(i, k) \in \boldsymbol{V}, i \neq k$ (ordered). A set $\boldsymbol{Z} \subset \boldsymbol{V} \backslash \{i, k\}$ statisfies the **back-door criterion** relative to (i, k) in $\mathcal{G}$, if:

- $\boldsymbol{Z} \cap \mathrm{De}(i) = \emptyset$, and
- $\boldsymbol{Z}$ blocks all *back-door paths* from i to k in $\mathcal{G}$, i.e., all paths between i and k that start with an arrow into i $(i \leftarrow \cdots k)$

Theorem Let $(\mathcal{G}, P)$ be a CBN. If $\boldsymbol{Z} \subset \boldsymbol{V}$ satisfies the **back-door criterion** relative to (i, k) in $\mathcal{G}$, we have

$$p(x_k | do(x_i)) = \int_{x_{\boldsymbol{Z}}} p(x_k | x_i, x_{\boldsymbol{Z}}) p(x_{\boldsymbol{Z}}) dx_{\boldsymbol{Z}}$$

The back-door criterion is **sufficient** for adjustment.

Proof in Notes for Lecture 5!

More identification strategies next week!

Credits

- **Recommended reading** is *Elements of Causal Inference* Chapter 6.6, and Chapter 3.1-3 in *Causality* (Pearl, 2009)
- This lecture, *builds upon and extends* the slides of the course "*Causality*" (2021) by Christina Heinze-Deml

Thank you!